

TECHNICAL NOTES

2020 CENSUS-BASED REGIONAL, PROVINCIAL/HIGHLY URBANIZED CITY, AND CITY/MUNICIPALITY LEVEL POPULATION PROJECTIONS

Regional Level Population Projections

The 2020 Census-Based Population Projections (CBPP) at the regional level adopted the Cohort-Component Method, the same method used for the national level 2020 CBPP. This method projects the future population based on a baseline population, fertility, mortality, and migration. Whereas the national population projections cover a 35-year horizon for years 2020 to 2055, the regional population projections only cover a 15-year span from years 2020 to 2035 due to the decreased reliability of subnational projections in more distant years.

Spreadsheets from the Population Analysis System (PAS) of the United States Census Bureau (USCB) were used to aid in the preparation of inputs and finalization of regional population projections. The Demographic Analysis and Population Projection System (DAPPS) software, also from the USCB, was utilized to calculate the preliminary regional population projections from 2020 to 2035.

Consistent with the national population projections, the official census counts from the 2020 Census of Population and Housing (CPH) were used as inputs for the base population at the regional level. These counts were adjusted to midyear population (01 July 2020).¹

Regional fertility data from various sources of Civil Registration and Vital Statistics, National Demographic Health Surveys, and the 2020 CPH were assessed to establish the baseline regional total fertility rate (TFR). Age-specific fertility rates were also adjusted for some regions due to irregular patterns while maintaining the baseline regional TFR. The TFR estimates for 2021, 2022, and for each five-year period at the regional level from 2025 to 2035 were projected while ensuring consistency with the national projected TFRs.² Projecting the regional TFRs aligned with the Scenario 2 required regional baseline TFRs, baseline national TFR, and the projected national TFRs. Similar to the previously approved methodology for the national population projections under Scenario 2, wherein the TFR will gradually decline from 2.1 in 2020 to 1.9 in 2021 until 2035, the regions will follow the same trend.

The 2020 age-specific death rates (ASDRs) from the national population projections were used as a model pattern of mortality along with the calculated level of infant mortality rate (IMR) per region to arrive at regional ASDRs and life expectancy at birth (LEB) by sex. With the calculated IMR per region and national 2020 ASDR from the national population projections, regional ASDRs, and baseline regional LEB were derived from the output life tables of the MATCH function of MortPak software. The baseline regional LEB were then used to project regional LEB. The projected regional LEB were used as input along with the baseline and projected national LEB estimates for 2020, 2021, 2022, 2025, 2030, and 2035 for population projections. The mortality

¹ The *PopAgeMove* function of DAPPS was utilized to move the population distribution to midyear.

² This was performed through the *PROJTFR32* worksheet of PAS.

inputs used for the regional population projections were the regional ASDRs by sex for the year 2020, and the projected LEB by sex from 2021 to 2035.

There were two inputs for migration – primary migration which is the regional net number of international migrants and secondary migration which is the projected interregional net migration rates (NMRs).

Primary migration was measured using data on the number of Filipino emigrants obtained from the Commission of Filipinos Overseas and number of immigrants retrieved from the Bureau of Immigration. To derive the regional net number of international migrants by age group and sex from the 2020 to 2035 as an input to be used for the primary migration in the projection of each region, the estimated 2020 net number of international migrants by age group and sex used in the generation of the national population projections were distributed to all regions following a proportionate allocation. The resulting age-sex structure for the 2020 net number of international migrants was assumed to be consistent until 2035.

Secondary migration utilized the 2010 CBPP projected NMRs for 2020 to 2035. The projected NMRs were for five-year periods (2015 to 2020, 2020 to 2025, 2025 to 2030, and 2030 to 2035) and were annualized to 2020, 2025, 2030 and 2035 by dividing the NMRs by five (5).

Provincial/Highly Urbanized City (HUC) Level Population Projections

The 2020 population projection round at the provincial and HUC levels adopted a top-down approach to ensure alignment with more reliable regional and national totals derived from censuses, surveys, and administrative data. As such, projections at the provincial and HUC levels were adjusted to match these higher-level figures. The base population for these projections was drawn from the 2020 CPH, covering the period from midyear 2020 to 2030 — a shorter horizon than the regional projections due to decreasing reliability over time. HUCs were treated as equivalent to provinces in compliance with PSA Board Resolution No. 1, Series of 2017-143, which mandates separate data generation for HUCs. For projection inputs, two sex-disaggregated province/HUC census counts were required. Proportions of provincial and HUC populations relative to their respective regions were calculated using the 2010 and 2020 CPH data. These proportions, along with census reference dates, were used to derive logistic growth rates for the projections. Projection of the province/HUC proportions to one or more future dates use the calculated logistic growth rates. Finally, the projected proportions were multiplied by the regional population projections to estimate the population at the province/HUC levels.³

The age-group distribution of province/HUC population projections by sex was estimated using the two-way raking process (also known as iterative proportional fitting and contingency table adjustment) to distribute the projected total populations of the provinces /HUCs into age groups by sex.⁴

³ This was performed using the SALGST worksheet of PAS.

⁴ This was performed using the CTBL32 worksheet of PAS.

City/Municipality Level Population Projections

The city/municipality level population projections will also be adjusted to be consistent with provincial-level totals, excluding HUCs, which are already included in the provincial projections. These projections use the 2020 CPH as the base population and cover the period from midyear 2020 to 2030. The methodology mirrors that of the province/HUC level, utilizing 2010 and 2020 census data to calculate each city/municipality's share of the provincial population. Logistic growth rates were derived from these proportions using the same formula applied at the provincial level. The projected proportions of the cities/municipalities relative to their provinces were multiplied by the provincial population projections to obtain the city/municipality population.

However, unlike the province/HUC level, the city/municipality projections did not use the PAS worksheet to generate age-sex distributions due to limitations on the number of cities/municipalities that can be processed. Instead, an interpolation process was used to distribute total population estimates into five-year age groups until 85 years and over by sex for each projection year through 2030.